How Memory Works

Psychologists study the mind as rational[2] and as a habit of the body, and both views explain part of how we **remember** what we read. Each view predicts something the other *misses*, which is why the *field keeps both* views in mind.

The brain holds about 10^{11} neurons and a signal crosses a synapse in 10^{-3} s, so a search costs $O(n \cdot d^2)$ steps for n items of length d. Numbers like these are rough, but they show the scale of the problem.

Models that are fine-tuned on a few examples forget less when they are fine-tuned again later, a result that holds for people learning a skill as well. Practice that is spread out leaves a stronger trace than practice crammed.

Some puzzles belong to the class of NP-complete problems, and no shortcut is known for them. Memory search is easier: a good cue narrows it at once, which is why cues matter so much.

We review frequently because a memory reviewed frequently is cheaper to recall, as the spacing studies show. The gain is largest when the gap between reviews grows a little each time it works.

The idea goes back to Alan Turing’s 1950 article “Com-

[2] We are not suggesting that people are irrational, only that habits matter as much as reasons do when we study.

puting Machinery and Intelligence," which asked whether machines can think.

Each later chapter returns to this question from another side, and each one ends with exercises that ask you to recall what it said, so the reading turns into practice at once.

A book starts a paragraph by indenting its first line instead of leaving a gap, and the lines after it go back to the margin as this one does.

[1] H. Ebbinghaus. Memory: A Contribution to Experimental Psychology, translated by H. Ruger and C. Bussenius, 1913.
[2] H. Roediger and J. Karpicke. Test-enhanced learning, 2006.
[3] N. Kornell and R. Bjork. Learning concepts and categories: is spacing the enemy of induction? 2008.